Example 4.60 How many diagonals are there in a polygon with n sides?

Solution:
A polygon of n sides has n vertices. By joining any two vertices of a polygon, we obtain either a side or a diagonal of the polygon. Number of line segments obtained by joining the vertices of a n sided polygon taken two at a time is ${}^nC_2 = \frac{n(n-1)}{2}$. Out of these lines, there are n sides of polygon. Therefore, number of diagonals of the polygon is

$$\frac{n(n-1)}{2} - n = \frac{n(n-3)}{2}.$$

In particular for a pentagon and heptagon (Septagon), number of diagonals respectively are $\frac{5(5-3)}{2} = 5$ and $\frac{7(7-3)}{2} = 14$.



Exercise - 4.3

1. If ${}^nC_{12} = {}^n C_9$ fin ${}^{21}C_n$.
2. If ${}^{15}C_{2r-1} = {}^{15} C_{2r+4}$, fin r.
3. If ${}^nP_r = 720$, and ${}^nC_r = 120$, fin n, r.
4. Prove that ${}^{15}C_3 + 2 \times {}^{15} C_4 + {}^{15} C_5 = {}^{17} C_5$.
5. Prove that ${}^{35}C_5 + \sum_{r=0}^{4} {}^{(39-r)}C_4 = {}^{40} C_5$.
6. If ${}^{(n+1)}C_8 : {}^{(n-3)} P_4 = 57 : 16$, fin the value of n.
7. Prove that ${}^{2n}C_n = \frac{2^n \times 1 \times 3 \times \cdots (2n-1)}{n!}$.
8. Prove that if $1 \leq r \leq n$ then $n \times {}^{(n-1)} C_{r-1} = (n-r+1){}^nC_{r-1}$.
9. (i) A Kabaddi coach has 14 players ready to play. How many different teams of 7 players could the coach put on the court?
   (ii) There are 15 persons in a party and if each 2 of them shakes hands with each other, how many handshakes happen in the party?
   (iii) How many chords can be drawn through 20 points on a circle?
   (iv) In a parking lot one hundred , one year old cars, are parked. Out of them f ve are to be chosen at random for to check its pollution devices. How many different set of f ve cars can be chosen?
   (v) How many ways can a team of 3 boys,2 girls and 1 transgender be selected from 5 boys, 4 girls and 2 transgenders?
10. Find the total number of subsets of a set with

    [Hint: ${}^nC_0 + {}^n C_1 + {}^n C_2 + \cdots + {}^n C_n = 2^n$]
    (i) 4 elements (ii) 5 elements (iii) n elements.
11. A trust has 25 members.
    (i) How many ways 3 off cers can be selected?
    (ii) In how many ways can a President, Vice President and a Secretary be selected?